NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY
GREATER NOIDA

(An Autonomous Institute Affiliated to Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT-BASED LEARNING (PBL) FINAL REPORT
MONTH 3 EVALUATION

MULTI-CRITERIA DEMAND-SUPPLY MATCHING & LOGISTICS OPTIMIZATION PLATFORM

A Unified Algorithmic Implementation of Classical Data Structures (Units 1 to 5):
Trees, Heaps, Graphs, Shortest Paths, Dynamic Programming, Branch & Bound, and B+ Trees

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DSA PBL (Data Structures & Algorithms)

CANDIDATE'S DECLARATION

I hereby declare that the work presented in this Project-Based Learning report entitled "Multi-Criteria Demand-Supply Matching & Logistics Optimization Platform", submitted to the Department of Computer Science and Engineering, Noida Institute of Engineering and Technology, Greater Noida, in partial fulfillment of the requirements for the course Data Structures and Algorithms (DSA PBL), is an authentic record of our own work carried out under the supervision of our faculty mentor.

The algorithmic implementations, empirical benchmark measurements, and analytical formulations presented in this report have been directly developed and executed by me. To the best of my knowledge, this report does not contain any fabricated data or undocumented claims.

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CERTIFICATE OF APPROVAL

This is to certify that the Project-Based Learning report entitled "Multi-Criteria Demand-Supply Matching & Logistics Optimization Platform", submitted by Tripti Verma, has been evaluated and approved as satisfying the academic requirements for the course Data Structures and Algorithms (DSA PBL).

The student has successfully developed, verified, tested, and documented the 8-phase algorithmic platform covering:

1. Unit 1: Trees, AVL Self-Balancing Trees & Binary Max-Heap Priority Queues
2. Unit 2: Graph Theory, Traversals (BFS, DFS), Minimum Spanning Trees (Kruskal, Prim with DSU), and Shortest Paths (Dijkstra, Bellman-Ford, Floyd-Warshall)
3. Unit 3: Dynamic Programming (0/1 Knapsack, LCS, Matrix Chain Multiplication, Resource Allocation)
4. Unit 4: Branch and Bound — Travelling Salesman Problem (TSP) with Matrix Reduction
5. Unit 5: Advanced Indexing — B+ Tree for Multi-Level Catalog Indexing and Range Queries
6. Integrated Matching Engine: Multi-criteria composite allocation framework.

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ABSTRACT

In modern distributed logistics, emergency humanitarian relief, and high-velocity supply chain networks, efficiently allocating constrained supplies to competing demand orders while adhering to multi-dimensional physical, monetary, and temporal constraints is a foundational computational challenge. Naive flat-file and brute-force approaches incur prohibitive polynomial or factorial time complexities, leading to catastrophic allocation delays and route inefficiencies during peak operational surges.

This project designs, engineers, benchmarks, and validates a unified, production-grade Multi-Criteria Demand-Supply Matching & Logistics Optimization Platform built strictly from foundational Data Structures and Algorithms without relying on black-box third-party dependencies.

The architecture synthesizes eight modular algorithmic phases aligned directly with the institutional DSA syllabus:

- Phase 1 (Unit 1): Self-Balancing AVL Trees (O(log N) price indexing) and Binary Max-Heaps (O(log N) urgency priority retrieval).
- Phase 2 (Unit 2): Versatile Graph representation (Adjacency List/Matrix), BFS/DFS traversals, and Connected Component cluster isolation in O(V + E) time.
- Phase 3 (Unit 2): Regional infrastructure backbone minimization using Kruskal's algorithm (with DSU path compression and union by rank) and Prim's algorithm, both converging to the identical optimal backbone cost of $168.00.
- Phase 4 (Unit 2): Shortest path navigation via Dijkstra's algorithm for standard transit routes, Bellman-Ford for routes featuring green delivery subsidies (negative weights) and negative-cycle detection, and Floyd-Warshall for all-pairs precomputed hub-to-hub dispatch.
- Phase 5 (Unit 3): Dynamic Programming formulations including 0/1 Knapsack for vehicle payload maximization, Longest Common Subsequence (LCS) for client-supplier contract specification alignment, Matrix Chain Multiplication (MCM) for multi-hop pipeline scheduling, and Resource Allocation DP for multi-center utility maximization.
- Phase 6: An integrated Multi-Criteria Matching Engine synthesizing price compatibility, specification similarity, and transit proximity into a composite scoring framework.
- Phase 7 (Unit 4 - Final Review Module): Travelling Salesman Problem (TSP) optimization using Least-Cost Branch and Bound (LCBB) with Matrix Reduction. Validated on standard 4-city ($80.00) and 5-city ($28.00) benchmark instances, and deployed on the 5-hub regional network to find the optimal closed delivery tour of $125.00 with 15 branches pruned.
- Phase 8 (Unit 5 - Final Review Module): Multi-level B+ Tree catalog indexing of order M=3, featuring automatic leaf and internal node splitting, O(log N) exact lookups, O(N) sequential leaf traversal, and O(log N + K) range querying.

All 26 automated unit and contract tests pass with 100% success rate in 3.53 milliseconds. The platform supports interactive CLI execution, local HTTP serving, and serverless cloud deployment on Vercel.

CHAPTER 1: INTRODUCTION & PROBLEM FORMULATION

1.1 Project Background & Industrial Context

Modern supply chain ecosystems—spanning emergency medical relief distribution, e-commerce fulfillment hubs, and heavy manufacturing procurement—face the critical computational challenge of allocating scarce supplies to competing demands under multi-dimensional constraints. Orders arrive with non-negotiable budget caps, urgency deadlines, and specialized qualitative tags (e.g., cold_chain, sterile, express, cleanroom). Suppliers maintain discrete inventory capacities, per-unit catalog prices, and fixed regional distribution warehouses. The physical transport grid introduces non-uniform transit costs, road congestion delays, and municipal green-delivery subsidies that produce negative transit costs.

1.2 Problem Statement & Computational Bottlenecks

A naive system that scans supplier databases linearly, processes orders in arrival order, and evaluates vehicle routes using brute-force permutations fails under real-world conditions:

1. Linear Catalog Search is O(N), inducing severe query latency during surge periods.
2. Unprioritized Order Ingestion risks starving critical life-saving demands behind routine bulk requests.
3. Graph Disconnection Blindness leads to dispatching vehicles across unreachable physical network clusters.
4. Sub-optimal Multi-Stop Dispatching causes delivery vehicles to make redundant round trips, multiplying fleet fuel and transit overheads.
5. Combinatorial Explosion in multi-stop routing (N! permutations) causes intractable calculation delays without branch pruning.

1.3 Project Objectives & Scope

The primary objectives of this Month 3 Final Milestone are:

- Maintain 100% backward compatibility with all working phases (Phases 1 through 6).
- Implement Unit 4: Travelling Salesman Problem using Least-Cost Branch and Bound (LCBB) with Matrix Reduction.
- Implement Unit 5: Functional B+ Tree of order M=3 for multi-level supplier catalog indexing, handling leaf/internal node splits, point lookups, and range queries.
- Execute automated testing and record actual empirical benchmarks without fabrication.
- Compile and deliver the complete consolidated Month 3 PBL Final Report in DOCX and PDF formats.

1.4 Syllabus Mapping to Platform Modules

Syllabus Unit	Core Data Structure / Algorithm	Platform Implementation Module
Unit 1	Self-Balancing AVL Trees & Binary	modules/phase1_structures.py

	Max-Heap	
Unit 2	Graph Representations, BFS, DFS, Connected Components	modules/phase2_graph.py
Unit 2 (cont.)	Minimum Spanning Trees (Kruskal with DSU, Prim)	modules/phase3_mst.py
Unit 2 (cont.)	Shortest Paths (Dijkstra, Bellman-Ford, Floyd-Warshall)	modules/phase4_shortest_path.py
Unit 3	Dynamic Programming (0/1 Knapsack, LCS, MCM, RAP)	modules/phase5_dp.py
Integration	Multi-Criteria Decision Analysis & Matching Engine	modules/phase6_matching_engine.py
Unit 4 (Final)	Branch & Bound: TSP with Matrix Reduction	modules/phase7_tsp_branch_bound.py
Unit 5 (Final)	Advanced Indexing: Multi-Level B+ Tree Catalog	modules/phase8_bplus_tree.py

CHAPTER 2: SYSTEM ARCHITECTURE & PIPELINE

The system employs a multi-stage pipeline designed around pure Python standard library structures. Incoming orders and supplier inventory records enter discrete indexing tiers:

1. Priority Tier: High-urgency demand orders are queued in a Binary Max-Heap, ensuring O(1) top-order retrieval.
2. Catalog Tier: Supplier catalog records are indexed into both an in-memory AVL Tree (keyed by price) and a B+ Tree of order M=3 for multi-level indexing and rapid range scanning.
3. Network Tier: The physical distribution network is modeled as a weighted Graph, precomputing reachability and all-pairs dispatch metrics via Floyd-Warshall.
4. Matching Tier: Top demands are matched against candidate suppliers within budget ceiling using LCS tag similarity, producing an overall composite compatibility score.
5. Fleet & Route Optimization Tier: Dispatch allocations are constrained by vehicle capacity using 0/1 Knapsack DP, and multi-stop delivery routes are scheduled using Least-Cost Branch and Bound TSP.

CHAPTER 3: FINAL REVIEW MODULES (UNITS 4 & 5)

3.1 Unit 4: Travelling Salesman Problem using Branch and Bound (Phase 7)

Unit 4 addresses the optimization of multi-stop delivery vehicle routes servicing multiple regional distribution hubs. The Travelling Salesman Problem (TSP) seeks a closed simple cycle visiting every specified hub exactly once with minimum total transit cost.

Mathematical Formulation of Least-Cost Branch and Bound (LCBB) with Matrix Reduction:

1. Matrix Reduction:

Given an N x N cost matrix C with diagonals set to infinity:

- Row Reduction: Subtract min_j C[i][j] from each row i, accumulating reduction cost R_rows.
- Column Reduction: Subtract min_i C[i][j] from each column j, accumulating reduction cost R_cols.
- Initial Lower Bound: LB_root = R_rows + R_cols.

2. State Space Tree Exploration:

Each node represents an assigned path prefix. When evaluating a transition from parent u to child v:

- In child matrix: set row u to infinity, column v to infinity, and C[v][start] to infinity (preventing premature subtours).
- Reduce the child matrix to determine incremental reduction cost r_child.
- Child Lower Bound: LB_child = LB_parent + parent.reduced_matrix[u][v] + r_child.

3. Branch Pruning:

If LB_child >= BestCost (the cost of the best valid complete tour discovered so far), the branch is pruned immediately, eliminating factorial subtree evaluations.

Empirical Verification Results for Unit 4:

- Benchmark 1 (4-City Standard Matrix): Verified optimal tour City_A -> City_B -> City_D -> City_C -> City_A with cost $80.00 (Initial LB: $70.00, 12 nodes explored, 5 branches pruned, runtime 0.1143 ms).
- Benchmark 2 (5-City Textbook Instance): Verified optimal tour Depot_1 -> Depot_4 -> Depot_2 -> Depot_5 -> Depot_3 -> Depot_1 with cost $28.00 (Initial LB: $25.00, 11 nodes explored, 6 branches pruned, runtime 0.1045 ms).
- Regional Logistics Application (5 Core Hubs): Deployed on North_Hub, Airport_Logistics_Center, Tech_Park, Central_Square, and East_District using Floyd-Warshall metric distances. Produced optimal closed tour North_Hub -> Airport_Logistics_Center -> Tech_Park -> Central_Square -> East_District -> North_Hub with total cost $125.00 (Initial LB: $100.00, 29 nodes explored, 15 branches pruned, runtime 0.3558 ms).

3.2 Unit 5: B+ Tree Catalog Indexing Engine (Phase 8)

Unit 5 implements a multi-level B+ Tree of order M=3 for supplier and product catalog indexing. Unlike standard binary search trees, the B+ Tree maintains all data records exclusively in leaf nodes while internal nodes store separator guide keys.

Structural Mechanics & Invariants:

- Order M = 3: Maximum M - 1 = 2 keys per node; a node splits upon accumulating M = 3 keys.

• Leaf Node Split: When a leaf overflows with 3 keys, it splits into left (1 key) and right (2 keys). The lowest key of the right leaf is copied UP to the parent as a separator key, and doubly-linked sibling pointers (next_leaf / prev_leaf) are updated.
• Internal Node Split: When an internal node overflows with 3 keys, the median key is pushed UP to the parent and excluded from both child internal nodes.
• Root Split: When the root overflows, a new internal root is allocated with one key and two children, increasing tree depth by 1.

Empirical Verification Results for Unit 5:
• Structural Split Verification: Sequential insertion of keys [50, 20, 80, 10, 30, 70, 90, 40, 60] produced a depth-3 tree with 3 internal nodes and 6 leaf nodes. Sequential leaf traversal yielded strictly sorted keys [10, 20, 30, 40, 50, 60, 70, 80, 90].
• Catalog Price Search: Exact lookup for unit price $88.00 retrieved Titan Heavy Foundries in 0.0019 ms.
• Procurement Range Query: Range query [$35.00, $120.00] traversed starting leaf and scanned sibling pointers, retrieving all 3 qualifying catalog records (FarmFresh $40.00, Titan $88.00, Helios $110.00) in 0.0047 ms in O(log N + K) time.

CHAPTER 4: EMPIRICAL BENCHMARKS & TEST VERIFICATION

The platform includes an automated unit test suite consisting of 26 tests across test_all.py and test_api.py. All 26 tests execute and pass in 3.53 milliseconds. The measured performance metrics across all phases are detailed below:

Phase & Module	Time Complexity	Space	Measured Time	Empirical Output / Invariant
Phase 1: AVL Tree Search	O(log N)	O(N)	0.0008 ms	Balanced BST (BF = 0)
Phase 1: Max-Heap Pop	O(log N)	O(N)	0.0012 ms	Strict priority order
Phase 2: BFS Traversal	O(V + E)	O(V)	0.0110 ms	10 nodes visited
Phase 2: DFS Traversal	O(V + E)	O(V)	0.0064 ms	Spanning tree created
Phase 3: Kruskal's MST	O(E log E)	O(V + E)	0.0326 ms	Cost: $168.00 (12 evaluated)
Phase 3: Prim's MST	O(E log V)	O(V + E)	0.0136 ms	Cost: $168.00 (12 evaluated)

Phase 4: Dijkstra Routing	O((V+E) log V)	O(V + E)	0.0185 ms	North_Hub -> Central: $22
Phase 4: Bellman-Ford	O(V * E)	O(V)	0.0241 ms	North_Hub -> Rural: $14
Phase 4: Floyd-Warshall	O(V^3)	O(V^2)	0.0482 ms	10x10 all-pairs matrix
Phase 5: 0/1 Knapsack DP	O(N * W)	O(N * W)	0.1270 ms	Max Utility: 1786.00
Phase 5: LCS Alignment	O(M * N)	O(M * N)	0.0084 ms	Spec similarity score
Phase 5: Matrix Chain DP	O(N^3)	O(N^2)	0.0110 ms	Optimal cost: 30,000
Phase 5: Resource Alloc.	O(M * K^2)	O(M * K)	0.0195 ms	Max Utility: 30.00
Phase 6: Multi-Criteria	Composite	O(D+S+V)	0.1437 ms	6 allocations fulfilled
Phase 7: TSP 4-City LCBB	O(2^N * N^2)	O(N * 2^N)	0.1143 ms	Cost: $80.00 (5 pruned)
Phase 7: TSP 5-City LCBB	O(2^N * N^2)	O(N * 2^N)	0.1045 ms	Cost: $28.00 (6 pruned)
Phase 7: TSP Logistics	O(2^N * N^2)	O(N * 2^N)	0.3558 ms	Cost: $125.00 (15 pruned)
Phase 8: B+ Tree Search	O(log_M N)	O(N)	0.0019 ms	Exact match found ($88)
Phase 8: B+ Tree Range	O(log_M N + K)	O(N)	0.0047 ms	3 records in [$35, $120]
Phase 8: B+ Tree Scan	O(N)	O(N)	0.0034 ms	Full leaf traversal

CHAPTER 5: CHALLENGES, LIMITATIONS & CONCLUSION

5.1 Engineering Challenges & Solutions

1. Duplicate Keys in Search Trees: Handled by attaching bucket lists to nodes rather than duplicate nodes.
2. Disjoint Graph Reachability: Handled by running connected components detection before shortest path calculations.
3. Negative Subsidies: Handled by pairing Dijkstra (for positive weights) with Bellman-Ford (for negative weights).
4. Combinatorial Explosion in Routing: Mitigated by Least-Cost Branch and Bound with Matrix Reduction.

5.2 Explicit System Limitations

1. Static Graph Topology: Edge transit costs are fixed; real-time dynamic congestion updates are not yet modeled.
2. In-Memory Persistence: Data structures reside in volatile memory without persistent disk serialization.
3. Advanced AI/ML Roadmaps: Deep learning embeddings (GNN) and reinforcement learning pricing remain future enhancements.

5.3 Conclusion

The project has successfully engineered, verified, and validated a pure Python multi-criteria demand-supply matching platform. With the integration of Unit 4 (TSP Branch & Bound) and Unit 5 (B+ Tree Indexing), all 5 institutional DSA syllabus units are fully operational and validated with 26 passing tests.